Chapter 2
Installing System 21 Explorer

Product Information

The software is available on CD-ROM.

This is what is shipped:

- System 21 Explorer. This contains the program software, including Designer, and the JBA System 21 Explorer Help.
- Administration Tools
  This contains programs which all appear separately on your Windows program list and can be run as standalone programs. You must either install all of them or none of them, they cannot be selectively installed:
  - System Configuration
  - User Role Menu Maintenance
  - Character Conversion
  - Window Recognition
  - Technical Reference
  - Currency Download
  - Email Server Agent.
- Samples.

Note: No Server objects are shipped with JBA Explorer or Explorer Designer.

Pre-Installation Considerations

Please take time to read this section before you start installing.

Authorisation Codes

You need to enter the following Authorisation Codes before using the software.

Software	Code
JBA System 21 Explorer	21/V3
JBA System 21 Explorer Designer	DG/V3

Hardware Requirements

You must have a hard disk with at least 36Mb available for stand-alone installation.

*Note: For a Network installation you need
46.4Mb for the Server and 18Mb for the Client.*

Software Requirements

AS/400 Server

You must have JBA System Manager Version 3.5.2 installed on the Server.

Warning: If TCP/IP is used to connect to an AS/400 Server, the TCP/IP licensed program (57xxTC1) must be installed on to the AS/400. Please note that this is automatically installed with Operating System OS/400 V3R7M0 or later

Note: The latest release of System Manager 3.5.2 must be installed on the server if you wish to use the Action Tracker or Action List functions.

UNIX Server

You must have the following software installed on the Server:

- JBA System 21 for AIX (inc. Version 3.5.2 applications)
- JBA Runtime Release 1.3.5 or later.

PC Client

Supported configuration (platform/router).

Authorisation for Explorer (21V3) and Designer (DGV3) as required.

Note: Please refer to the latest JBA International System 21 Supported Configuration Information sheet for pre-requisites and current version details. This sheet is available on the JBA BID system.

Warning: Any site not using a supported configuration will require JBA Branch agreement for local support of the product. Please confirm the setup is supported BEFORE installing the product.

All previous Server objects are now shipped with System 21 Application Manager or with Guidelines 3.4 Server runtime support. Both of these are detailed on the System 21 Configuration Sheet, and are supplied with their own installation instructions.

About ODBC

If you plan to do either of these things:

- Use the Currency Download facility in Explorer
- Use Constructor to Activate a Business Process.

then you need to consider your ODBC setup when installing Explorer.

Note: Whatever you set up for Explorer is the basis if you go on to install Constructor.

Your decision about whether or not you need to consider ODBC will affect what you do during the Explorer installation.

If You Do Not Need To Consider ODBC

There are two things to note:

- You do not need to install ODBC.
- During the Explorer installation you will be presented with a check box called Register ODBC Dependent Files. You leave this unchecked.

If You Need To Consider ODBC

There are things you need to do before and during the Explorer installation.

Before Installing Explorer

You need to install the ODBC driver manager: you must have ODBC32 level 2, or above, installed on the system. This is typically installed by products such as Microsoft Office.

You can do this to check the version of an ODBC installation:

- Launch the ODBC applet from Control Panel, or the relevant folder.
- Click the About tab and check the version reported under Administrator.
- If the major version is lower than 2, or if no About tab is shown, you must update your ODBC software using that supplied with this shipment pack. If this applet is not present you will need to install it.

To install the basic ODBC Administration Tools a Setup program has been supplied on the distribution CD in the subfolder \ODBC. To install run the Setup.exe program from the installation directory.

During The Explorer Installation

You will be presented with a check box called Register ODBC Dependent Files. You need to check this box.

What this does is to register the following components:

```
com0013.exe
com0010.dll
MSRDO20.dll
JBAParms.dll
```

These components must be registered in order to support Explorer's Currency Download function and the Activation function in Constructor.

Note: These files are always copied onto the installation directory during an installation but they are registered only when you check the box.

If you check the Register ODBC Dependent Files check box, but have not installed the ODBC driver manager, you can continue but error messages are displayed when the installer tries to register the above files.

If you have not registered the ODBC Dependent Files you will not be able to use the Currency Download or Activate a Business Process.

If you forget to check the Register ODBC Dependent Files check box, or, you decide later that you do need to consider ODBC then you can register the files manually. You do not need to reinstall Explorer or Constructor. This is what you do:

- Use the regsvr32 command to register the .DLL files.
- Register the EXE files by running them from a command line.

About Designer And System Configuration

JBA Explorer Designer is an integral part of Explorer. When you install Explorer you also install Designer. However, Designer is a tool for the System Administrator so please take care to disable permissions for general users.

System Configuration is a separately installed module within the Administration Tools. If you install this module then System Configuration appears as a separate program on your Windows program list. It can also be run from expconfg.exe.

You can access Designer only if the Designer field, in the Preferences Window, has been checked. Use System Configuration to enable the Designer field in the Preferences Window.

You can access System Configuration by using the System Configuration button on the Preferences Window. If this button is also disabled then you need to reset permissions using System Configuration as a standalone program rather than as part of Explorer.

About Technical Reference

Program Files

The Technical Reference program files are shipped as part of Explorer's Administration Tools, which gets installed as part of the Explorer Installation procedure.

You uninstall them as part of the Explorer uninstallation procedure.

Databases

The Technical Reference databases have to be purchased separately and are available on CD-ROM.

This is what is shipped:

- Generic Database
- Style Database
- Drinks Database.

There is no installation or uninstallation procedure for the Technical Reference Databases. You can use the databases directly from the CD or copy them manually to your PC. If you have copied the databases to your PC then you can manually delete them.

About Internet Explorer And Document Viewer

If you want to use the Document Viewer facility then please make sure that you have installed Microsoft Internet Explorer version 3 or above. This is a Web browser which the Document Viewer uses to view Web pages.

About Fonts

When you install Explorer the fonts are automatically installed.

The installation puts the fonts into C:\Program Files\JBASys21\Fonts and then copies them into C:\Windows\Fonts.

There are three available fonts:

- Arial Monospaced
- Arial Monospaced CE
- Arial Monospaced Zero-Dot.

The font to be used by Explorer is determined by the Font= statement in the JBALU2 section of jbasmg.ini. You can enter any of the three available fonts, plus those covered by Japanese support.

To change the fonts you need to manually change a line in jbamsg.ini configuration file. For example:

```
Font=10.Arial Monospaced CE
```

You can enter any font, but only the three fonts included in the [FONTS] section of jbamsg.ini are supported. If you add another font manually, you must also put a record in the [FONTS] section, especially if it is a Symbol type font.

In Explorer you can use a bold override setting, using the Preferences Screen Window. This updates the BoldFont= setting in jbasmg.ini from 0 to 1.

They are saved as .TTF files in the installed software directory with a subdirectory \Fonts.

Note: For foreign languages, you can get alternative fonts separately, and amend the INI file attribute to search for that font first.

Usually these fonts install correctly. However, you may get an error during the installation:

```
Arial Monospaced Font Not Found
```

This is caused by a font having been previously uninstalled incorrectly. The installation process then cannot install the new font as it expects to and reports that the font is not found.

The way around this problem is to manually install the font to the Fonts directory. To do this you copy them from C:\Program Files\JBASys21\Fonts into C:\Windows\Fonts.

An incorrectly installed font also causes problems with screen presentation.

Tip: When you have finished the installation, and rebooted, it is a good idea to check that the fonts have been installed.

For System 21 Explorer, within jbamsg.ini, in the [JBAS21DEFAULTS] section, change the line:

```
ChangeGUIFont=0
```

To:

`ChangeGUIFont=1`

This forces explorer to use the same font as the emulator (as set using Font=). This is primarily used for Japanese, but is also useful when you want to use the Arial Monospaced CE font.

For more information on the jbamsg.ini file, see the Tips and Techniques section of this help file.

About The Email Server Agent

Warning: The Email Server Agent is designed to run unattended and should be installed onto a Network Server rather than a users machine.

Warning The Work Management application must be installed on your server if you want to use the Email Server Agent.

The Email Server Agent forwards email messages generated by the Work Management Engine. It periodically asks Explorer to tell it all the Activity List items for the role *EMAIL. The Email Server Agent executes each activity in turn, sending the text that would have been shown on the screen as a MAPI compliant email.

This is what you do:

Step 1 Install Explorer

When installing Explorer please take the Administration Tools option. Administration Tools includes the Email Server Agent.

Step 2 Install Email System

Before using the Email Server Agent you must install a MAPI-compliant email system such as Microsoft Exchange or Windows Messaging.

For Windows NT

The Email Server Agent uses Microsoft Active Messaging version 1.1.

For Windows 95

The Email Server Agent needs a version of Microsoft Exchange greater than 4.0 in order to support Active Messaging.

If you have Microsoft Exchange version 4.0 or lower then please do one of the following:

- Install the DLL’s from the Exchange 4.0 Server release rather than the original Exchange 4.0 client released with Windows 95.
- Install any version of Exchange greater than 4.0.
- Download the explorer installation program, EXUPDUSA.EXE from ftp://ftp.microsoft.com/softlib/MSLFILES/.

Note: This is effectively an Exchange 4.0 Service Pack. First make a complete backup, then run the update.

The Explorer installation program tries to register the active messaging component olemsg32.dll. This fails under Windows 95 if your version of Microsoft Exchange does not support Active Messaging.

After you have upgraded Microsoft Exchange, you must manually register Olemsg32.dll using the REGSVR32 command. If Olemsg32.dll has not been registered correctly, you get a Failure To Create MAPI Session Error when starting the Email Server Agent.

Step 3 Start Explorer

Start Explorer from your Program List.

Select the Email Role Menu.

Tip: If it does not exist then create a new role menu that has a role code of *EMAIL. Please refer to the Tutorial in the User Role Maintenance section in the Explorer help guide.

Step 4 Start Email Server Agent

Start the Email Server Agent from your Program List to display the Email Server Agent Window.

Select the Start button.

Warning: If you want to shut down the Work Management server, to which Explorer is connected, then also close the Email Server Agent and close Explorer.

About Currency Conversion And Download

Currency Conversion

This is an Explorer Toolbar option and which you can enable using the System Configuration utility.

Select the toolbar button to enable the conversion tooltip feature. This means that when you are in a System 21 application you can position the mouse pointer over a field and press Ctrl + F1. This displays a tooltip with the conversion calculation and results.

Note: When a Currency Conversion is requested the Currency Calculator window is displayed, minimised. The Currency Calculator allows the Explorer user to perform ad hoc calculations, change conversion settings and set up the standard conversion.

The Currency Conversion function retrieves its currency information from a local access database JBAEMU.mdb, the location of which can be specified using the System Configuration utility. This database contains a copy of the relevant System 21 financial files required to perform currency conversion.

Warning: This local database must be populated from the financial files on the server before the conversion can be used. The local database should also be refreshed if currency data on the server is amended, as for a change in currency rates.

Currency Download

This is the utility you use to refresh the local database. This is installed as part of the Administration Tools and appears as a separate item on your Program List.

The utility uses ODBC to communicate with the server. Please refer to the About ODBC help section.

When downloading, the ODBC datasource you select must specify the server files location. For example, if using an AS/400 as the System 21 server then the data source used to download the currency data should have the library which contains the currency files FLP081 and FLP083 in its list of default libraries.

About The Japanese Installation

Note: You can now run Explorer in Japanese. The PC files needed to make Explorer work in Japanese are always shipped with the English version.

Warning: Explorer is Japanese enabled - not DBCS enabled. It is not currently possible to run Explorer in any DBCS environment other than Japanese.

Note: If you want to install the Japanese version, first follow the instructions to install the English version.

Installing The Software (English)

Warning: Please read the Pre-Installation Considerations section before you start.

Note: If you have any part of a previous version of JBA System 21 Explorer on your machine then you need to follow the instructions for Upgrading The Software.

Note: Refer to the EXPREADME.TXT file included on the installation CD for details.

This is what to do:

Step 1

Run Setup.exe from the installation directory to start the installation.

Tip: Online help and instructions are provided during the installation. It is most likely that you will want to accept all the default settings presented to you and need only click on the Next button each time in order to progress through the installation.

Step 2 - Setup Information Window

The readme file is displayed for you to view.

Warning: Please make sure you have read the About ODBC help in the Pre-Installation Considerations section.

There is a check box called Register ODBC Dependent Files. Check this box only if you plan to do either of these things:

- Use the Currency Download facility in Explorer
- Use Constructor to Activate a Business Process.

Step 3 - Standalone/Server Window

You choose to perform either a standalone installation, for your PC only, or a network installation on the server.

Warning: Please refer to the section called About Network Installations.

Step 4 - Select Components Window

You can choose to install these components:

- Explorer: this contains the program software, including Designer, and the JBA System 21 Explorer Help.

- Essentials: It contains essential system files which contribute to the calculation of the space used by the installation. It is included for you with your selection where appropriate.
- Administration Tools: This contains programs which you can run standalone and appear separately on your Windows program list. You must install all of them or none of them, they cannot be selectively installed:
  - System Configuration
  - User Role Menu Maintenance
  - Character Conversion
  - Window Recognition
  - Technical Reference
  - Currency Download. Warning: Please refer to the help sections, called About ODBC and Currency Conversion And Download, in the Pre-Installation Considerations section
  - Email Server Agent. Warning: Please refer to the help section, called About The Email Server Agent, in the Pre-Installation Considerations section.
- Samples.

You can view the amount of space used by each part of the installation.

You can choose to enter your own destination directory. The default directory is:

```
c:\Program Files\JBASys21
```

The installation software checks the contents of the directory you enter here, but no other directory, to see if there is a version of Explorer remaining. If it finds a version then it displays this warning message:

```
Setup has detected JBA System 21 Explorer in this directory. Please uninstall the existing
version before you proceed.
```

It also looks for Administration Tools and displays a similar message.

Step 5 - Select Program Folder Window

You can enter your own name for the folders in your program list. The default names given to you are JBA System 21 and JBA Help.

A summary of all your selected components is displayed for you to confirm the list before completing the installation. If you are not happy with the summary then click the Back button as many times as you need and amend any of your installation entries.

Step 6

At the end of the installation you can choose to reboot now or later.

Warning: JBA strongly recommend you reboot now. You must reboot before installing Constructor.

Step 7

At the end of the installation, if you have chosen to install all the components, you will have these folders in your program list:

JBA Help

- Technical Information Help
- Technical Reference Help
- JBA System 21 Explorer Help.

JBA System 21

- JBA System 21 Explorer
- System Configuration
- User Role Maintenance
- Character Conversion
- Window Recognition
- Technical Reference
- Currency Download
- Email Server Agent.

You will also have a new directory:

`c:\Program Files\JBASys21\ExpSamples`

containing the samples.

Note: You can use System Configuration in Explorer to set up the servers. Please use the System Configuration Window.

Installing The Software (Japanese)

Note: If you want to install the Japanese version first you need to install the latest English version of Explorer.

This is what to do:

Step 1

Install the latest English version of Explorer. Please refer to Installing The Software (English).

Step 2

Reboot the PC to make sure that the English text does not remain in memory.

Step 3

Take a copy of jbasmg.ini, and save for future reference.

Step 4

Using Notepad, or any other windows based text editor, open both the following files: jbasmg.ini and Jp.ini. The default location for these files is:

`C:\Program Files\JbaSys21.`

Step 5

Change the English Text in jbasmg.ini to Japanese by copying from Jp.Ini.

Note: Ignore the text for the countries [AGB], [CAB] and [USB].

Note: The ,16 after the title is the font size used on the splash screen.

Step 6

Change the line:

```
TranslationLanguage=eng
```

to:

```
TranslationLanguage=jpn
```

This sets the extension for the language files which are loaded at runtime.

Step 7

Change or add this line in the [JBAS21DEFAULTS] section

```
ChangeGUIFont=1
```

This results in Explorer using the specified font for all windows.

Step 8

Change or add these lines to the [JBAS21DEFAULTS] section:

```
DBCSAToEFile=C:\Program Files\JBASys21\Pcsxlt.aed
DBCSEToAFile=C:\Program Files\JBASys21\Pcsxlt.ead
```

This tells Explorer where to find the DBCS ASCII EBCDIC conversion tables.

Step 9

Configure a Japanese AS/400.

Tip: For full details, please refer to Character Translation.

Here is an example of how to do this, using JAPANES as the name of the AS/400:

```
[Hosts]
JAPANES=AS/400 TCP/IP
[JAPANES]
DBCS=1
TranslationFile=C:\Program Files\JbaSys21\Jbadtab.dat
```

Step 10

Save the INI file.

Step 11

Make sure that the AS/400 has:

- Japanese OS/400 installed
- System 21 installed
- TCP/IP running.

Step 12

You can now start Explorer.

Upgrading The Software (Full Upgrade)

Note: If this is the first time that JBA System 21 Explorer has been installed on your machine then you need to follow the instructions for Installing The Software.

Note: Refer to the ExpREADME.TXT file included on the installation CD for details.

Note: If your installation CD specifically says PIP then you need to follow the instructions for Upgrading The Software (PIP Upgrade).

This is what to do:

Step 1 Backup Work Files

Note: When you install Explorer the installation software keeps a list of any work files which it has installed. This list is then used by the Uninstallation software and any files on the list are deleted. This means that any files you amend are deleted during the Uninstallation but any files you create will remain. However, JBA strongly recommends that you backup all your files.

Copy away any work files and extracts you have created and want to use again. Look for these first in the default pathway which is:

`C:\Program Files\JBASys21\.`

These may include:

- New or changed Role Menu files (*.RMU).
- Modified Software Recognition data contained in the Designer table in the JBARTime.mdb Access database.
- User and Role authorisation data, contained in the Roles table in the JBARTime.mdb Access database.
- New or changed (via Constructor) activity/business process information contained in JBADTime.mdb Access database.

Step 2

Uninstall any previously installed components. Refer to Uninstalling The Software.

*Note: Uninstall reminds you that it has not deleted the directory containing *.RMU files. It is referring to files that you have created. Delete them now if you do not want them, otherwise they will remain on file.*

Step 3

Warning: Please make sure that you reboot after uninstalling.

If you have defined data then once the upgrade is complete you can merge your existing data into the new database by running the MergeRuntimeDatabases application called MergeRTime.exe.

Step 4

You are now ready to install the new software. See Installing The Software.

Upgrading The Software (PIP Upgrade)

Note: A PIP upgrade is a short term improvement issued by JBA. It will be followed by a full upgrade. Unless your installation disk states specifically that it is a PIP then follow the instructions for a Full Upgrade instead.

Warning: Make sure you have the correct PIP upgrade version in relation to your currently installed version. Refer to the EXP352v.TXT file included on the installation CD for details (where v is the version identifier).

This is what to do:

Step 1

You may need to copy away any work files and extracts that you have previously created and want to use again. Look at the PIP disk to see what has been supplied. Only copy away those files and extracts which will be overwritten by those supplied on the PIP.

Look for the existing ones in the default pathway which is:

`C:\Program Files\JBASys21\.`

These may include:

- New or changed Role Menu files (*.RMU)
- Modified Software Recognition data contained in the Designer table in the JBARTime.mdb Access database
- User and Role authorisation data, contained in the Roles table in the JBARTime.mdb Access database
- New or changed (via Constructor) activity or business process information contained in JBADTime.mdb Access database.

Step 2

From the PC, make sure that Windows has been started.

Step 3

Look on your CD for the EXP352*v* directory (where v is the version identifier) and from here run Setup.exe to start the installation. The changed objects are in subdirectory called ExpUpdate.

Step 4

Online help and instructions are provided during the installation. It is most likely that you will want to accept all the default settings presented to you and need only click on the Next button each time in order to progress through the installation.

The readme file is displayed.

Warning: Please make sure you have read the About ODBC help in the Pre-Installation

Considerations section.

There is a check box called Register ODBC Dependent Files. Check this box only if you plan to do either of these things:

- Use the Currency Download facility in Explorer
- Use Constructor to Activate a Business Process.

You can also choose the destination location. The screen displays the location of System 21 Explorer on your PC. You may change this location using Browse but the system checks to see if there is a version of System 21 Explorer in this directory and if it cannot be found it displays the message:

```
Setup could not find JBA System 21 Explorer in this directory
```

and changes the location back again.

Step 5

At the end of the installation you can choose to reboot now or later.

Warning: JBA strongly recommends you to reboot now. If you are going on to upgrade Constructor please reboot first. You will not need to reboot after installing Constructor.

Step 6

Note that if you have defined data then once the upgrade is complete you can merge your existing data into the new database by running the MergeRuntimeDatabases application called MergeRTime.exe.

Warning: If you subsequently uninstall Explorer please be aware that the whole of Explorer will be uninstalled, not just the PIP Upgrade.

Uninstalling The Software

Note: You need to do this before you reinstall or do a full upgrade (not a PIP Upgrade) of JBA System 21 Explorer.

Warning: If you uninstall after a PIP Upgrade then the whole of Explorer will be uninstalled not just the PIP.

This is what to do:

Step 1

Make sure that the Explorer Data Exchange Area application (ExpCloud.exe) is not running. This application is initiated by Explorer but remains active until your machine is rebooted or you manually end the task

Step 2

Click on Start\Settings\Control Panel and then click on the Add/Remove Programs icon. This is a standard Windows application.

All your currently installed software is listed for you.

Select JBA System 21 Explorer and click the Add/Remove button and Explorer will be uninstalled for you.

Step 3

Warning: Please make sure that you reboot after uninstalling. You must reboot before installing Explorer.

Step 4

When you have rebooted it is a good idea to check that the fonts have all been uninstalled and if not then delete them manually. They are saved as .TTF files in the installed software directory with a subdirectory \Fonts.

About Network Installations

Warning: For a Network Client Installation Explorer must be installed from the server. Do not install directly from supplied media or your main directory. The Network Client Installation provides access to server installed products.

A Network Installation is where you load the software onto a Server, minimally load the application onto the PC, then run the application against the server.

- Network Server - loads Server portion. You can run client installations from here. Network versions are run against this.
- Network Client - loads minimal Client portion. You can install from Server install. Provides access to Server Installed products.

Note: The performance of the Network Version depends upon the speed and quality of the network. The time to install applications will vary depending on the speed of your Server, your Network and your PC.

The advantages of a Network Installation are:

- You install just once, and everyone can install from the central place.
- Once installed on a server, the PC Client install takes a fraction of the normal installation time.
- The Network Administrator can setup configuration files on their PC, copy them to the server, and they will be auto-installed onto all other PCs during a Client Install. So jobs such as Server configuration, preference setting, Server screen recognition changes from standard only need to be carried out once.
- Space saving on the PC (see Pre-Requisites).

Tip: Please refer also to Lueinfo.cfg.

Tip: For a fast install onto many PCs, run a Network Server Install once only, followed by short Network Client installation onto each PC. These PCs will then run Explorer over a network.

Using A Silent And Automatic Installation

Read this if you want to install Explorer onto a number of PCs. By using a Silent or Automatic Installation you can significantly reduce the amount of time taken to complete the task, because these types of installation do not require user input.

Note: You cannot use a Silent or Automatic Installation for Constructor.

Silent Installations

Instead of entering all the installation details when prompted, as you would for a normal installation, you put all your responses into a Silent Response File and let the software pick them up from there. You receive no prompts and enter no data and so because of this the installation is faster.

The format of response files resembles that of an .INI file. A response file is a plain text file consisting of sections containing data entries.

No messages are displayed during a Silent Installation. Instead, a log file named Setup.log captures installation information, including whether the installation was successful. You can review the log file and determine the result of the installation. Any error messages that would normally be displayed are written to the Errors.log file.

There are two stages involved:

Firstly, create the Silent Response File. You can do this either by setting one up from scratch or by modifying one of the samples issued by JBA. JBA recommends that you use the second method where possible.

Secondly, use the Silent Response File on any number of PCs to run a Silent Installation.

Stage 1 Create A Silent Response File

Run Setup.exe with the -r option.

You enter data and perform what appears to be a normal installation however, in this case the software creates a Silent Response File at the same time, called Setup.iss.

Setup.iss now contains all your responses and is placed in the Windows folder.

You can then use this file for silent installations on other PCs.

Note: JBA supply sample response files and recommend that you modify one of these rather than creating one from scratch.

Stage 2 Run A Silent Installation

Run Setup.exe with the -s option.

This time the installation takes its responses from Setup.iss.

Use the -f1 and -f2 parameters to specify the name and location of the response file and the location of the Setup.Log file. See the Command Line arguments section.

To verify if a silent installation succeeded, look at the ResultCode value in the [ResponseResult] section of Setup.log. An appropriate return value is written after the ResultCode keyname:

- If the resultcode = 0 then the installation succeeded
- If the resultcode is anything other than zero then the installation failed. You can check the Errors.log for details of why the installation failed

Automatic Installations

The automatic installation gives you more control over the installation than the Silent Installation. You receive no prompts and enter no data, however, you get a visual interface, so that you can monitor the progress of the installation. Also, installation stops at the Finish/Reboot Window to give you the chance to reboot now or later.

Run Setup.exe with the -a option.

The automatic installation gets its responses from the Automatic Response File. There is no way of generating this file, you must use one of the sample files with the appropriate settings.

Use the -f3 parameter to specify the name of the response file. If this parameter is omitted then setup will look for Setup.isa in the same directory as Setup.exe.

No messages are displayed, instead any error messages are written to the Errors.log file, if any errors have been recorded then you are notified and referred to the log file.

Command Line Arguments

There are two ways of running Setup.exe with command line arguments:

Method 1

You can type in the command line parameters at the command prompt. For example:

```
Setup.exe -s -f1C:\Mydir\setup.iss -f2C:\Mydir\Mydir.log
```

Method 2

You can add the arguments to the Setup.ini and run the Setup.exe as usual.

The following is an example of a typical Setup.ini file:

```
[Startup]
AppName=JBA System 21 Explorer
FreeDiskSpace=484
EnableLangDlg=Y
CmdLine=-s -f1C:\Mydir\setup.iss -f2C:\Mydir\Mydir.log
```

For automatic installs you must add the command line parameters to the Setup.ini file.

The command line parameters specified in Setup.ini are read first, and then any command line parameters passed to Setup.exe from the command prompt are appended.

Examples

The following examples illustrate the use of Setup.exe, including use of the command line parameters -a, -s, -f1, -f2 and -f3:

`setup`	Launches Setup and tries to load Setup.ins from the same directory that contains Setup.exe.
`setup -s`	Launches setup in silent mode and tries to load Setup.ins and Setup.iss from the folder containing Setup.exe. If the -f1 parameter is not used when running in silent mode, Setup looks for the response file Setup.iss in the same folder as Setup.exe. A log file is created in the same folder. However, if you are running the installation from a CD then this will not be possible.
`setup -s -f1C:\Mydir\Mydir.iss`	Launches setup in silent mode and tries to use Mydir.iss (from the C:\Mydir folder) as the response file. This example also creates the log file Setup.log in the same folder as that of the response file (C:\Mydir). The log file has the default name Setup.log if the -f2 parameter is not provided along with -f1.
`setup -s -f1C:\Mydir\Mydir.iss - f2 C:\Mydir\Mydir.log`	Launches setup in silent mode, uses Mydir.iss from the C:\Mydir folder, and generates the log file Mydir.log in the C:\Mydir folder.
`setup -a`	Launches setup in Automatic mode and tries to load Setup.isa from the folder containing Setup.exe. Any messages which would normally be displayed, with the exception of Fatal errors will be written to the Errors.log file.
`setup -a -f3C:\Mydir\Mydir.isa`	Launches setup in Automatic mode, tries to use Mydir.isa (from the C:\Mydir folder) as the response file. Any messages which would normally be displayed, with the exception of Fatal errors will be written to the Errors.log file.

Notes

Do not leave a space between command line parameters and options.

Setup.exe command line parameters and options are not case sensitive, with the exception of the SMS parameter.

Do not use the Automatic and Silent parameters together:

- For an Automatic Installation use the -a and -f3 parameters
- For a Silent Installation use the -s, -f1 and -f2 parameters.

You can use the -SMS parameter with both types of installation.

The -SMS parameter prevents a network connection and Setup.exe from closing before the installation is complete. This parameter works with installations originating from a Windows NT server over a network. Please note that SMS must be uppercase; this is a case-sensitive parameter.

Setup Log File

Setup.log is the default name for the silent installation log file, and its default location is the same folder as Setup.ins. You can specify a different name and location for Setup.log using the -f1 and -f2 parameters with Setup.exe. You must do this when installing from a CD because the setup cannot write to the CD drive.

The [ResponseResult] section, contains the result code indicating whether or not the silent installation succeeded. An integer value is assigned to the ResultCode keyname in the [ResponseResult] section. The return value is placed after the ResultCode keyname:

The Setup.log file for a successful silent installation is as follows:

```
[InstallShield Silent]
Version=v5.00.000
File=Log File
[Application]
Name=JBA System 21 Explorer
Version=3.5.2e
Company=JBA
[ResponseResult]
ResultCode=0
```

Using Sample Response Files

Warning: If using any of these sample files you must make sure that the installation path is correct, that is the appropriate drive letter is used.

JBA supply these sample Response Files for both Silent and Automatic Installations:

Client.iss and Client.isa	Default selections that are offered to the user during the normal Client installation.
Default.iss and Default.isa	The default selections offered during a normal Standalone installation.
Network.iss and Network.isa	The default selections offered during a normal Network installation.
Pip.iss and Pip.isa	The default selections offered during a normal pip upgrade installation.
Lite.iss and Lite.isa	The default selections offered during a normal Lite installation.

All the files are commented telling you where to make changes. Only modify the sections that are commented.

Using A Silent Uninstall

Note: You can run a silent uninstallation regardless of whether the installation method used was Silent, Automatic or Normal.

When you run a Silent Uninstallation, the uninstallation software checks for any shared files. However, unlike a normal uninstallation, where you would expect to see a message asking whether you want to remove a shared file, instead, this message is not displayed. The software does not remove the shared file but it reduces the reference count to zero. This is equivalent to a normal uninstallation where you would select the No To All option when this window first appears.

Files

Before running the software check that you have the following two files:

An Uninstallation Executable File: IsUninst.exe

This file is automatically installed in the Windows folder.

IsUninst.exe can be run from the command line like any other executable file, but it is usually run from the Add/Remove Programs Properties sheet.

For silent uninstallation you must run IsUninst.exe from the command line.

An Uninstallation Log File: .ISU

During the installation an uninstallation log file is created for each setup. The uninstallation log file contains a record of all the uninstallation related events that occur during the installation.

The log file is installed in the installation directory.

The name of the ISU file depends on the application being installed.

Command Line Arguments

When passing a path and file name in a command line parameter, you must enclose a name containing one or more spaces in double quotation marks in order to avoid misinterpretation.

```
-f<log file name>
```

Specifies the location and name of the uninstallation log file.

For example:

```
-f"C:\Program Files\JBASys21\JBAExplorer.isu"
```

This will run the uninstallation in the same way as if you had run it from the Add/Remove Programs Properties sheet.

```
-a
```

Specifies that the uninstallation should run in silent mode.

For example:

```
-a -f"C:\Program Files\JBASys21\JBAExplorer.isu"
```

This will run the uninstallation in silent mode, there is no notification that the uninstallation is complete. It is recommended that you reboot your machine after uninstalling.

Configuring The PC Software

The Installation software automatically configures the PC as much as is possible, or prompts with advice about what steps are required. The details in this section cover areas definitely not automatically configured by the Installer.

PC Configuration (Not Host Specific)

Warning: Most settings in the jbasmg.ini file should be maintained through Explorer rather than manually.

To configure the INI file use System Configuration. This is installed as part of Administration Tools and is a separate program on your Windows program list. This means that after installation you have the option to use System Configuration within Explorer or as a standalone program.

The purpose of System Configuration is to give permissions to the person using Explorer by enabling or disabling various facilities for them. You can also use it to set up the servers.

Non-English Operating Systems And Changed Sign On Panels

If the AS/400 Sign On Panel has the first words something other than Sign On, then extra processing is required before Explorer is run. This normally arises due to the Operating System being Non-English, or because the Sign On panel has been customised. The Translated or Customised values for the emulator defaults in Explorer must be changed in the jbasmg.ini file. Please refer to the SelectHost section of jbasmg.ini for further details.

PC Configuration (Using UNIX And NT)

This section is targeted for use by a competent UNIX or NT technician. Please refer to the accompanying Tips and Techniques, pre-requisite software and any relevant UNIX or NT documentation material.

Much of the required configuration for UNIX and NT use may have already taken place during the installation of any pre-requisites.

However, the next sections are a full checklist of requirements for Server and PC Client.

UNIX Or NT Host Requirements.

Unless otherwise specified, you must be logged in as root to perform all UNIX or NT tasks.

1. Create a new UNIX or NT user to run System 21/UNIX/NT and set the profile up as a normal System 21 user.

The user should not be asked for an input during login and must login straight to the UNIX/ksh command line. Therefore the user ID should NOT run any sort of menu system.

2. Edit the TCP/IP Services (/etc/services) file so that it contains the following line:

```
jqserv24000/tcp# JBA System 21 / UNIX
jbachp24500/tcp# JBA System 21 / UNIX
```

Replace 24000 and 24500 with unique port numbers for use by inetd on the system. To make sure that this port is not already used, get your System Administrator to check the /etc/services file and also run netstat -an for any other services which use this same number.

3. Edit the TCP/IP Configuration (/etc/inetd.conf) file so that it contains the following lines:

```
jqserv stream tcp nowait s21user / JBA_openroute/bin/sviewstart sviewstart
jbachp stream tcp nowait s21user / JBA_openroute/bin/sviewstart sviewstart -c
```

Replace S21user with the name of the dedicated System 21 user ID on UNIX.

4. Run the following commands to force the INETD daemon to recognise the above changes:

```
inetimp  (skip this command if using UNIX version 4 or above)
refresh -s inetd
```

Check that the svsignon script in your Release directory has SUID-root switched on by running the following commands:

```
chown root /Release/*/exports/*/bin/svsignon
chmod ug+s /Release/*/exports/*/bin/svsignon
```

UNIX Load Balance

This is not supported in the current version of the software.

Defining UNIX Configuration

You can define a UNIX Configuration for both the PC and the Server.

PC

In the System 21 directory the jbasmg.ini file shipped with Explorer holds the Server information together with a server name (the section entitled [JBAVT]). The server name defaults to jqserv. You can change this to a suitable name.

In the TCP/IP directory the \etc\services file is the TCP/IP Services Configuration file and relates the servers with a unique port number (for example, 24000/TCP) and a description.

Note: The location of the services file may vary between PCs, so if it is not in the stated location then please search the drive for the required file.

The application server - JQSERV can be changed to a suitable name.

In both cases, you can also change the port numbers if needed.

Host

In the TCP/IP directory the file /etc/services is a duplicate file to the TCP/IP Services file on the PC. The server and port numbers must match those on the PC services file.

Note: The location of the services file may vary between PCs, so if it is not in the stated location then please search the drive for the required file.

In the TCP/IP directory the file /etc/inetd.conf is the task driver for the application, and resides on the Server in the same directory as the Server services. It is accessed by port number from the Server services only, and holds the path and task details.

For multiple databases, please use a new port number and server name throughout the setup.

Setting Up Interfaces

You can use System 21 Explorer to set up interfaces between Explorer and external programs.

To help you with this JBA provide sample programs. When you install Explorer you have the option to install the samples to a Samples directory.

Adding User Toolbar Buttons

You can set up your own buttons on the Explorer Toolbar and then use the button to execute an ActiveX component or a program.

To do this, select Add from the System Configuration Toolbar Window to display the Toolbar Button Details window.

Select the Type panel and choose to start either an object or a program.

Then select the Action panel:

- If you are starting a program then only the Program field is enabled.
- If you are starting an object then the Server and Class fields are enabled.

Note: The Class you define must contain a method called Execute. This method must accept an integer as a parameter. The execute method is invoked each time the button is pressed. The integer parameter identifies which button has been pressed.

You might want to use this sample:

```
Samples\ToolbarUserButtons\Button.vbp
```

Calling Before And After

From Designer you can call some programming function before or after or both before and after you display a System 21 application window. You might want to do this to insert your own program or screen design.

To do this, click the Details button from the Designer Panel Window to display the Panel Details Window.

Here you enter the details of the ActiveX component. The Server is the name of the ActiveX component and Class is the name of the class within the ActiveX component which contains the appropriate methods (BeforeExecute and AfterExecute) called from Designer.

Fill in the appropriate text boxes on this form. The boxes you complete depend on whether the ActiveX component is called before you display the window, or after, or both.

Note: For the sample ActiveX, Server is OleCallingTest and Class is clsCallingTest.

Call Before and Call After are properties for a given panel or screen:

- Call Before starts an ActiveX component when the panel is about to be shown or when the panel has been reset.
- Call After starts an ActiveX component when any key relating to the emulator has been pressed. This is a key that transmits the panel back to the Server. For example: F Keys, Control Key (which resets the panel), Page Up, Page Down, Return.

These ActiveX components can be used in various ways such as, to hide panels, set information into panels, confirm keypresses, and alter keypresses.

You might want to use this sample: Samples\DesignerCallBeforAfter\CallingTest.Vbp

Notes About Creating A Call ActiveX

In your ActiveX component, there is only one requirement for each type of call, without which it will not work. For a Call Before there needs to be a BeforeExecute method and for a Call After there needs to be an AfterExecute method. In both methods three parameters are passed in, these represent:

- – The emulator (Lue)
- – The Show property of the panel
- – The KeyID reference.

In a new project add a class and a BAS module.

In the module add the essential Main method. For example:

```
Public Sub Main()

End Sub
```

For a Call Before, add to the class, a method called BeforeExecute with three passed in parameters, an object, a boolean and an integer. For example:

```
Public Sub BeforeExecute(objLue As Object, ShowFlag As Boolean, KeyID As Integer)

End Sub
```

For a Call After, add to the class, a method called AfterExecute with parameters, an object, a boolean and an integer. For example:

```
Public Sub AfterExecute(objLue As Object, ShowFlag As Boolean, KeyID As Integer)
End Sub
```

Within each method add the appropriate code, such as that to launch forms in the project.

As references are passed in for the Lue (as an object), the panel itself can be changed with the appropriate Lue methods.

The boolean passed in is more relevant in the Call Before than the Call After. If it is changed to false, then the emulator will not be visible.

The integer passed in is more relevant in the Call After than the Call Before. It identifies the key that has been pressed: this value can be changed by the method. For example a Return can be transformed into F3.

Adding Custom Buttons

From Designer you can set up your own buttons on the window you are reformatting and then use the button to execute an ActiveX component.

To do this, click the Add Button button from the Designer Panel Window to display Designer Button Details Window.

Define a Server and Class for the object to be executed by the button. The Server is the name of the ActiveX component. The Class is the name of a class, within that ActiveX component, that contains the Execute method.

Define the position of the button. The position is verified and if it overlaps another field, you cannot place the button and the OK button is disabled.

You might want to use this sample:

`Samples\DesignerCustomButtons\Oletest.Vbp.`

Notes About Creating A Call ActiveX

Make sure that the StartUp Object in the Project Properties is set to Sub Main.

Add a module to the project, and add the Main method.

Add a class to the project.

In the class, create a method called Execute, as this will be the method called from the custom button. For example:

```
Public Sub Execute(objLue As Object)
    frmMain.Show
End Sub
```

Add your code, within this method, to activate forms within the ActiveX component, Windows Applications and Other ActiveX components.

Note that an object is passed into the Execute method. This is a reference to the Emulator and this reference can be used to alter the Emulator Panel or Fields.

Create another method called ScreenChange, this is another method called from Designer. For example:

```
Public Sub ScreenChange(MagicNumber As String)
    gstrMagic = MagicNumber
End Sub
```

From this method, the ActiveX component will know when the panel (that launched the ActiveX component) has changed or has been exited. So code could be placed here to unload the ActiveX component should the MagicNumber change from the first pass of this method.

Communicating Externally With Explorer/Session Manager

There are two primary methods that can be used to externally communicate with Explorer/Session Manager. You use either OLE or DDE.

Note: OLE can also be referred to as ActiveX.

OLE Communications

Note: JBA supplies you with the test application ExternalCommsTest.vbp

You can use three methods when calling into Explorer/Session Manager:

- RunServerTask
- GetSessionData
- StickyFields.

RunServerTask Method

Syntax:

```
RunServerTask  TaskCode, TitleofTask, [ConnectToSystem], [UserID], [OverrideCompany],
[RunAttributes]
```

This is used to launch a task via Explorer. TaskCode and TitleofTask are mandatory parameters, the rest are optional.

GetSessionData Method

Syntax:

```
GetSessionData  SessionNumber, DataToRetrieve, ReturnedData
```

This method can be used to retrieve information about an active session.

ReturnedData will bring back the information that has been requested.

Set SessionNumber to zero to default to the last used session.

DataToRetrieve

Company

Company that the session is running in. (String)

EmulatorFont

Font that the session is using.

Format : Size.FontName. (String)

Environment

Environment that the session is running in. (String)

HostUserID

UserID used to start the session. (String)

KeyboardID

Returns the keyboard code for the session. (String)

Lu2Handle

A reference to the Lu2 HandleObject which allows manipulation of the session fields and function keys. (Object)

MagicNumber

Returns the magic number of the session. (String)

RunAttributes

Returns the run attributes of the current session, as defined in the Advanced properties panel for tasks in Explorer. (Long)

System

System the session is running on. (String)

SystemType

Type of system that the selected session is running on. (String)

TaskCode

Task Code for the selected session. (String)

TaskTitle

Task Title as it appears in the form caption of the session. (String)

Return Codes

These are possible return codes that GetSessionData can bring back:

0 means a successful retrieval of information.

-1 means no active tasks for the requested session number.

-2 means cannot find information requested.

StickyFields Method

Syntax:

```
StickyFields  FieldAction, FieldName, FieldValue
```

This is used to manipulate and alter the values in the sticky fields of a session.

FieldAction

0 means read the contents of a sticky field

1 means write to the contents of a sticky field

FieldValue

When reading the contents of a sticky field it will return the value for you, however when writing to a sticky field this variable needs to contain the information that you are intending to write.

Return Codes

These are possible return codes that StickyFields can bring back.

0 means Success.

-1 means could not attach to the Data Exchange Area (cloud).

-2 means error reading from a sticky field.

DDE Communications

Tip: JBA supplies four example macros:
- RunServerTask Template.txt
- GetSessionData Template.txt
- ManipulateSession Template.txt
- StickyFields Template.txt

There are also two demo macros:
- ManipulateSessionDemo.txt
- StickyFieldsDemo.txt

A DDE linked application can issue four commands to Explorer/Session Manager:

– RunServerTask

– GetSessionData

– ManipulateSession

– StickyFields.

These examples will clarify the syntax of the commands. Overall structure of an entire macro can been seen in the examples. In this document channel number always refers to the channel on which the DDE conversation is taking place.

<u>RunServerTask Command</u>

Syntax:

```
DDEExecute (channel number, “RunServerTask:  details”)
```

Details

Can be constructed from up to six parameters, however only the first two in the following list are mandatory, the rest are optional.

Note: the parameters must appear in this order: Task Code, Task Title, Connect to System, User ID, Override with Company, Run Attributes

Task Code

The code that identifies the task

Task Title

The caption that appears on the form running the task

Connect to System

Used to override the default system

UserID

Used to override the default UserID

OverrideWithCompany

Used to override the default company

RunAttributes

This is a number that specifies advanced properties of the task

GetSessionData Command

Syntax:

```
DDEExecute (channel number, "GetSessionData:  details")
```

Details

This is constructed as follows: Session Number; DataToRetrieve

SessionNumber

The number of the session that you want to retrieve the data from, zero can be used to specify the last used active session.

DataToRetrieve

Company

Company that the session is running in.

EmulatorFont

Font that the session is using.

Format : Size.FontName

Environment

Environment that the session is running in.

HostUserID

UserID used to start the session.

KeyboardID

Returns the keyboard code for the session

MagicNumber

Returns the magic number of the session.

RunAttributes

Returns the run attributes of the Long current session, as defined in the Advanced properties panel for tasks in Explorer.

System

System the session is running on.

SystemType

Type of system that the selected session is running on.

TaskCode

Task Code for the selected session.

TaskTitle

Task Title as it appears in the form caption of the session.

ManipulateSession

Syntax:

```
DDEExecute (channel number, “ManipulateSession:  read, details”)
DDEExecute (channel number, “ManipulateSession:  write, details”)
DDEExecute (channel number, “ManipulateSession:  keypress, details”)
```

ManipulateSession can carry out three distinct functions, hence why there are three possible versions of syntax.

For ManipulateSession: read, details is constructed as follows:

```
SessionNumber, FieldNumber, FieldType
```

For ManipulateSession: write, details is constructed as follows:

```
SessionNumber, FieldNumber, FieldType, FieldLength, TextToWrite
```

For ManipulateSession: keypress, details is constructed as follows:

```
SessionNumber, Keycode
```

SessionNumber

The number of the session that you wish to manipulate, zero can be used to specify the last used active session.

FieldType

There are two possible field types, input fields - such as text boxes, and output fields - such as labels. Input fields = 1, Output fields = 2.

FieldNumber

Specifies the number of the field, of the type you wish to manipulate. Note: the field count is zero based so the first input field and output field are always zero.

FieldLength

Only used when writing to a field. Specifies the length of the field that is being written to.

TextToWrite

The text that you want to insert into the field

KeyCode

This refers to the key that you are pressing in the session. F1-F24 have keycodes 43-64 respectively. Other useful codes include Enter (42), Page Up (73) and Page Down (74).

StickyFields

Syntax:

```
DDEExecute (channel number, “StickyFields:  details”)
```

Details are constructed as follows:

Action, FieldName, [Text to Insert]

Action

You can either write to or read from a sticky field. Insert an integer here to identify the action you wish to take. Read from field=0, Write to field=1.

FieldName

Name of the sticky field that you are manipulating.

Text to Insert

Only used when writing to a sticky field, do not include when reading from a field.

Troubleshooting

If you have any problems or questions about JBA System 21 Explorer then please refer to Tips And Techniques. This gives you:

- Troubleshooting help.
- Checklists.
- Detailed background information on various subjects such as the routers, the error codes, the configuration files shipped with the product.
- Useful suggestions for making further use of the product. For example how to carry out a trace.

Tips and Techniques refers to all these three interfaces, unless an individual interface is named specifically within the text:

  - System 21 Client Workspace
  - System 21 Explorer
  - System 21 Constructor.

Please refer to your JBA representative at JBA Product Support.